

Aitken acceleration in Norma: two formulations

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Abstract

This note records the two Aitken-acceleration formulations implemented in Norma’s non-overlapping Schwarz alternating solver, together with the displacement-form kinematic recovery they use. The purpose is to fix the exact formulas and index conventions so they can be reproduced in Sierra/SM. Both formulations are exposed through the multi-domain controller: the recursive Irons–Tuck variant (`relaxation: aitken`) and the non-recursive secant variant (`relaxation: aitken secant`). They are mathematically equivalent in exact arithmetic but differ in their initialization bookkeeping and in how they treat the interface velocity and acceleration.

1 Setting and fixed-point reformulation

Consider a domain decomposition of Ω into two non-overlapping subdomains Ω_1 and Ω_2 sharing the interface Γ . Norma drives the coupling *multiplicatively* (Gauss–Seidel ordering of the subdomain solves) with a strongly imposed Dirichlet condition on the Dirichlet side of the interface and a traction (Neumann) condition on the other. Let $\mathbf{g}$ denote the interface unknown, i.e. the displacement datum imposed on the interface. Following Sambataro and Tezaur [1] we write the multiplicative Schwarz sweep as a fixed-point map

$$\mathbf{g} \mapsto T(\mathbf{g}), \quad (1)$$

where $T(\mathbf{g})$ is obtained by solving the subdomains in order with the current interface data and projecting the resulting trace back onto Γ ; the converged interface state satisfies $\mathbf{g}^\star = T(\mathbf{g}^\star)$.

Define the *interface jump* (the fixed-point residual)

$$\mathbf{r}^{(n)} := T(\mathbf{g}^{(n)}) - \mathbf{g}^{(n)}. \quad (2)$$

In the Norma implementation (`src/schwarz.jl`) the two terms map to the code variables as

$$\mathbf{g}^{(n)} = \text{lambda_disp (from iteration } n-1), \quad T(\mathbf{g}^{(n)}) = \text{interp_disp},$$

so that $\mathbf{r}^{(n)} = \text{interp_disp} - \text{lambda_disp}$. This is the displacement mismatch the relaxation acts on; it is non-degenerate because `interp_disp` is the freshly projected neighbor trace, whereas the strongly imposed continuity makes the *post-imposition* interface residual identically zero on the Dirichlet side.

The relaxed update common to all variants is

$$\mathbf{g}^{(n+1)} = \mathbf{g}^{(n)} + \omega^{(n)} \mathbf{r}^{(n)} = \omega^{(n)} T(\mathbf{g}^{(n)}) + (1 - \omega^{(n)}) \mathbf{g}^{(n)}, \quad (3)$$

with relaxation factor $\omega^{(n)}$. Classical relaxation uses a fixed $\omega^{(n)} \equiv \omega$; the Aitken variants choose $\omega^{(n)}$ dynamically.

2 Version 1: recursive Irons–Tuck Aitken (relaxation: aitken)

With $\delta^{(n)} := \mathbf{r}^{(n)} - \mathbf{r}^{(n-1)}$, the Irons–Tuck recursion [3] sets

$$\omega^{(n)} = -\omega^{(n-1)} \frac{\mathbf{r}^{(n-1)} \cdot \delta^{(n)}}{\|\delta^{(n)}\|^2}. \quad (4)$$

The factor carries the damping history through the multiplicative $\omega^{(n-1)}$, which requires the iterate to have advanced by exactly $\omega^{(n-1)}\mathbf{r}^{(n-1)}$ for the recursion to remain consistent (see Section 4).

3 Version 2: non-recursive secant Aitken (relaxation: aitken secant)

This is the form derived in the original references, Sambataro–Tezaur [1] (Eq. 9) and Deparis–Discacciati–Quarteroni [2]. With $\mathbf{d}^{(n)} := \mathbf{g}^{(n)} - \mathbf{g}^{(n-1)}$ (the change in the interface iterate) and $\delta^{(n)}$ as above,

$$\omega^{(n)} = -\frac{\mathbf{d}^{(n)} \cdot \delta^{(n)}}{\|\delta^{(n)}\|^2} = \arg \min_{\omega} \|\mathbf{d}^{(n)} + \omega \delta^{(n)}\|^2. \quad (5)$$

The numerator uses the change of the iterate $\mathbf{d}^{(n)}$, not the previous residual, and there is no recursive $\omega^{(n-1)}$ factor. Norma forms $\mathbf{d}^{(n)} = \mathbf{g}^{(n)} - \mathbf{g}^{(n-1)}$ directly from the two stored interface iterates rather than from the recursion.

4 Relationship between the two forms

The update (3) gives $\mathbf{g}^{(n)} = \mathbf{g}^{(n-1)} + \omega^{(n-1)}\mathbf{r}^{(n-1)}$, hence

$$\mathbf{d}^{(n)} = \mathbf{g}^{(n)} - \mathbf{g}^{(n-1)} = \omega^{(n-1)}\mathbf{r}^{(n-1)}. \quad (6)$$

Substituting (6) into the secant factor (5) reproduces the Irons–Tuck factor (4) exactly. The two forms therefore coincide *whenever the interface iterate advances by exactly $\omega^{(n-1)}\mathbf{r}^{(n-1)}$* . They diverge only where that identity is broken by the initialization and warm-up bookkeeping (Section 5): the secant form reads the actual stored iterates $\mathbf{g}^{(n)}, \mathbf{g}^{(n-1)}$ and is robust to it, while the recursion silently inherits any inconsistency in the stored $\omega^{(n-1)}$ and $\mathbf{r}^{(n-1)}$.

5 Initialization

Let $\omega^{(1)}$ be a user-supplied initial factor (Norma: `relaxation parameter`, default 1) and N_0 a warm-up count (Norma: `aitken NO parameter`, default 1). For $1 \leq n < N_0$ the factor is held at $\omega^{(n)} = \omega^{(1)}$; the first dynamic factor is computed at $n = N_0$, once two residuals (Version 1) and additionally two iterates (Version 2) are on record.

6 Numerical safeguards

A single safeguard is retained: a floor on $\|\delta^{(n)}\|^2$ (10^{-20}) that guards the division in (4) and (5) when successive residuals are essentially identical; in that case the previous factor is reused (Version 1) or $\omega^{(1)}$ is used (Version 2). No value clamp is applied to $\omega^{(n)}$: clamping caps the very large or near-zero steps that produce the acceleration and was found to slow convergence, so it is omitted.

7 Which variable to relax: kinematic consistency

For dynamics the interface carries displacement $\mathbf{u}$, velocity $\mathbf{v}$, and acceleration $\mathbf{a}$. Under Newmark integration with a *fixed* step history $(\mathbf{u}_n, \mathbf{v}_n, \mathbf{a}_n)$ and parameters $\beta, \gamma, \Delta t$, the end-of-step kinematics lie on a one-parameter family,

$$\mathbf{u}_{n+1} = \mathbf{u}_{\text{pre}} + \beta \Delta t^2 \mathbf{a}_{n+1}, \quad \mathbf{v}_{n+1} = \mathbf{v}_{\text{pre}} + \gamma \Delta t \mathbf{a}_{n+1}, \quad (7)$$

with predictors $\mathbf{u}_{\text{pre}} = \mathbf{u}_n + \Delta t \mathbf{v}_n + (\frac{1}{2} - \beta) \Delta t^2 \mathbf{a}_n$ and $\mathbf{v}_{\text{pre}} = \mathbf{v}_n + (1 - \gamma) \Delta t \mathbf{a}_n$. Thus only one of $(\mathbf{u}, \mathbf{v}, \mathbf{a})$ is independent: the integrator's primary solve variable. The relaxation should act on that variable, with the remaining two recovered from (7).

Norma (displacement-form Newmark). The solver unknown is $\mathbf{u}$ (`src/time_integrator.jl, correct`). The secant variant therefore relaxes *displacement only*, $\mathbf{u}^{(n+1)} = \mathbf{g}^{(n+1)}$ from (3), and recovers

$$\mathbf{a}^{(n+1)} = \frac{\mathbf{u}^{(n+1)} - \mathbf{u}_{\text{pre}}}{\beta \Delta t^2}, \quad \mathbf{v}^{(n+1)} = \mathbf{v}_{\text{pre}} + \gamma \Delta t \mathbf{a}^{(n+1)}. \quad (8)$$

The legacy **relaxation**: `aitken` variant instead relaxes $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{a}$ independently with the same $\omega^{(n)}$. A ω -blend of two kinematically consistent triples is again consistent, so the two treatments agree when the incoming triple lies on (7); they differ slightly otherwise because Norma interpolates the three interface histories independently in time. Recovering from a single relaxed variable keeps the imposed interface state exactly on the integrator manifold.

Sierra/SM (acceleration-form). In an acceleration-form integrator the primary unknown is $\mathbf{a}$. The same principle then prescribes relaxing the *acceleration* and recovering $\mathbf{u}$ and $\mathbf{v}$ by inverting (7), i.e. replacing the role of $\mathbf{u}$ in (8) by $\mathbf{a}$. This is the relevant choice for reproduction in Sierra/SM.

8 Summary for reproduction

Formulation	Relaxation factor	Norma key	Variable relaxed
Irons–Tuck (recursive)	$\omega^{(n)} = -\omega^{(n-1)} \frac{\mathbf{r}^{(n-1)} \cdot \boldsymbol{\delta}^{(n)}}{\ \boldsymbol{\delta}^{(n)}\ ^2}$	<code>aitken</code>	$\mathbf{u}, \mathbf{v}, \mathbf{a}$ (same ω)
Secant (non-recursive)	$\omega^{(n)} = -\frac{\mathbf{d}^{(n)} \cdot \boldsymbol{\delta}^{(n)}}{\ \boldsymbol{\delta}^{(n)}\ ^2}$	<code>aitken secant</code>	primary unknown; recover rest

Table 1: The two Aitken formulations implemented in Norma. Here $\mathbf{r}^{(n)} = T(\mathbf{g}^{(n)}) - \mathbf{g}^{(n)}$, $\boldsymbol{\delta}^{(n)} = \mathbf{r}^{(n)} - \mathbf{r}^{(n-1)}$, and $\mathbf{d}^{(n)} = \mathbf{g}^{(n)} - \mathbf{g}^{(n-1)}$. For Norma's displacement-form Newmark the primary unknown is $\mathbf{u}$; for an acceleration-form integrator (Sierra/SM) it is $\mathbf{a}$.

On a representative dynamic non-overlapping Dirichlet–Neumann cantilever problem the secant variant converged in 2 Schwarz iterations per time step, versus 4 to 5 for the recursive variant, while both matched the monolithic single-domain reference to a relative tip error of $\sim 10^{-10}$.

References

- [1] G. Sambataro and I. Tezaur, *On the role of relaxation and acceleration in the non-overlapping Schwarz alternating method for coupling*, arXiv:2603.17143.

- [2] S. Deparis, M. Discacciati, and A. Quarteroni, *A domain decomposition framework for fluid-structure interaction problems*, MOX Report 45, Politecnico di Milano.
- [3] B. M. Irons and R. C. Tuck, *A version of the Aitken accelerator for computer iteration*, Int. J. Numer. Methods Eng. **1** (1969) 275–277.